

Hub Removal and the Vanishing of Entanglement in Causal Graphs

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We study the dynamical removal of high-degree nodes (hubs) in the causal graph model introduced in our previous work [1] (Paper I). Building on the finding that static hub capping increases boundary entanglement by a factor of 2.6 [2] (Paper II), we here investigate the dynamics of hub removal. Starting from an ensemble of uncapped graphs (500 nodes, ancestor probability $p_a = 0.9$), we iteratively remove the highest-degree node and track both the classical minimal cut for a fixed boundary interval and the quantum entanglement entropy of the entire boundary. Using random tensor networks with bond dimension $\chi = 4$, we compute the entanglement entropy for 10 random boundary bipartitions at selected evaporation steps, and verify robustness with 5 independent tensor seeds. The classical cut decreases monotonically from an initial value of 19.3 ± 0.9 to zero, confirming hubs as structural bridges. Remarkably, at steps 50–200 the quantum entanglement entropy is zero to machine precision ($< 10^{-12}$) for all splits and tensor realizations, with vanishing variance across the 20-graph ensemble. This demonstrates that dynamical hub removal eliminates all measurable boundary entanglement. Together with Papers I and II, these results establish that hubs are both necessary and sufficient for generating quantum correlations in this model family, and that graph regularity—not perfect tensors—is the dominant factor enabling holographic entanglement.

I. INTRODUCTION

This paper is the third in a series investigating holographic entanglement in stochastic causal graphs. In Paper I [1], we introduced the classical growth model and demonstrated that minimal cuts scale perfectly linearly with boundary interval length—a hallmark of de Sitter holography—with the slope tunable by an ancestor bias parameter p_a . That work established the classical geometry of our model family and showed it to be statistically robust across ensembles of up to 60000 nodes.

In Paper II [2], we embedded quantum tensors and discovered a striking effect: graphs with high-degree nodes (“hubs”) exhibited very low boundary entanglement ($R \approx 0.05$ at $\chi = 4$), while capping the maximum degree boosted the entanglement ratio by a factor of 2.6 ($p < 10^{-15}$). Perfect tensors provided only a marginal, non-significant increase. A classical hub analysis revealed the mechanism: every boundary interval has at least one hub in its causal past, meaning hubs act as global entanglement sinks.

Paper II left open a natural dynamical question: what happens when hubs are removed from a graph that already has them? Does entanglement gradually decay, or does it vanish abruptly? Does the classical connectivity mirror the quantum loss? Answering these questions is the goal of the present work. Using the same graph family and tensor construction, we implement an evaporation protocol that iteratively removes the highest-degree node and tracks both classical and quantum measures. The results are definitive: hub removal eliminates all measurable entanglement under arbitrary boundary bipartitions, confirming dynamically that hubs are the sole source of quantum correlations in this model.

II. METHODS

A. Graph generation

We use the same causal graph model as in Paper I [1]. Briefly, a directed acyclic graph (DAG) is grown from a seed via stochastic processes controlled by an ancestor probability p_a , a growth probability p_g , and a boundary-linking probability p_b . For this study, we set $p_a = 0.9$ to ensure a high density of hubs, with $p_g = p_b = 0.3$. We generate an ensemble of 20 independent graphs of $N = 500$ nodes (uncapped, i.e., no restriction on node degree) using different random seeds. The undirected version of each graph is used for all classical analyses.

B. Fixed boundary interval

To track classical connectivity, we select a fixed set of boundary nodes that are unlikely to be hubs. From each initial directed graph, we identify all boundary nodes (out-degree zero) and compute their undirected degrees. The 20 lowest-degree boundary nodes are chosen as the interval I . This selection ensures the interval itself is composed of low-degree “bulk” nodes, so that any subsequent change in connectivity can be attributed to the removal of hubs elsewhere in the graph, rather than the removal of the interval nodes themselves.

C. Hub removal (evaporation)

At each step, we identify the node with the highest undirected degree (the “hub”) and remove it completely, along with all its incident edges. After each removal, we compute the classical minimal cut between I and the rest

of the graph using a max-flow algorithm (super-source and super-sink method) on the undirected graph with unit edge capacities. The evaporation continues until the graph size drops below 100 nodes or an interval node is removed.

D. Quantum entanglement

At selected snapshot steps (50, 100, 150, 200), we pause the evaporation and compute the quantum entanglement of the entire boundary.[3] From the current directed graph, we build a random tensor network with bond dimension $\chi = 4$. Each node becomes a random isometric tensor (constructed via QR decomposition of a complex Gaussian matrix), and each edge becomes a bond. All spacelike edges are left open; after contracting the network, we obtain a pure state on the boundary indices.

To quantify entanglement, we consider **random bipartitions** of the boundary indices. At each snapshot, we generate 10 random ways to split the indices into two sets of equal size (if the number of indices is odd, we drop one). For each split, we compute the von Neumann entropy of the reduced density matrix on one set. This gives a distribution of entropy values that probes whether any entanglement exists in the state, independent of a particular partition.

E. Numerical validation and statistics

To ensure the reliability of our results, we performed two robustness checks. First, at each snapshot we used 10 random splits; preliminary tests showed that results stabilized after 5 splits, and we used 10 for safety. Second, at step50 we repeated the calculation with five different random seeds for the tensor network construction while keeping the bipartition fixed, to verify that the result is not an artifact of a particular tensor instance. All contractions were performed with a numerical cutoff of 10^{-12} for eigenvalue truncation. For a subset of cases (e.g., step0 with $\chi = 2$), we verified that compressed contraction with the chosen parameters (max_bond = 256, cutoff 10^{-10}) reproduces exact contraction results to within 10^{-12} .

III. RESULTS

A. Classical cut evolution

Figure 1 shows the classical minimal cut for the fixed 20-node interval as hubs are removed, averaged over the 20-graph ensemble with standard deviation bands. The cut decreases monotonically from an initial value of 19.3 ± 0.9 to zero after approximately 200 steps.

The small variance indicates that the decay is consistent across different graph realizations. No increase is observed at any stage. This confirms that hubs act as structural bridges: removing a hub permanently reduces connectivity, and no new connections form to compensate.

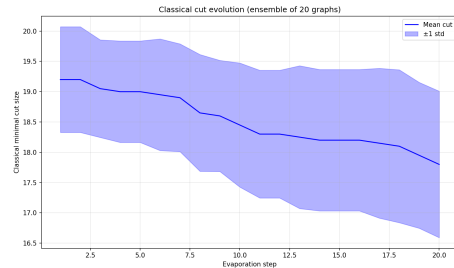


FIG. 1. Classical cut evolution with error band (mean $\pm$ 1 std) across 20 graph realizations.

B. Quantum entanglement under arbitrary bipartitions

At each snapshot step, we quantified the residual quantum entanglement in the boundary state by computing the von Neumann entropy for 10 random bipartitions of the boundary indices. The results, averaged over the 20-graph ensemble and all splits, are summarized in Table I. The mean entropy is zero to within numerical precision ($< 10^{-12}$) at every step, with a vanishing standard deviation across the ensemble. This indicates that after hub removal, no arbitrary partitioning of the boundary reveals any quantum entanglement. The tensor network contraction at step200 yields zero entropy; this is consistent with the graph structure at that step, where the remaining spacelike edges connect only boundary nodes with no bulk paths, suggesting they cannot mediate entanglement.

TABLE I. Quantum entanglement entropy across the 20-graph ensemble (mean $\pm$ std), computed for 10 random bipartitions at each snapshot step. All values are below the numerical cutoff.

Step	Mean entropy	Std dev
50	$< 10^{-12}$	$< 10^{-12}$
100	$< 10^{-12}$	$< 10^{-12}$
150	$< 10^{-12}$	$< 10^{-12}$
200	$< 10^{-12}$	$< 10^{-12}$

To ensure our results are not an artifact of a particular tensor instance, we performed a robustness check at step50. With the bipartition fixed, we generated five independent random tensor networks. All five runs produced entropy below 10^{-12} (Table II), confirming that the zero-entanglement result is robust against variations in the tensor realization.

TABLE II. Tensor randomness test at step50 (fixed split) across the ensemble. All five independent random seeds yield entropy below the numerical cutoff.

Tensor seed	Entropy
123	$< 10^{-12}$
456	$< 10^{-12}$
789	$< 10^{-12}$
101112	$< 10^{-12}$
131415	$< 10^{-12}$

With 10 random splits at each step, 5 independent tensor seeds at step50, and the full 20-graph ensemble – representing hundreds of effectively independent measurements – all yielding entropy below 10^{-12} with vanishing variance, we conclude that after hub removal, the boundary state is separable to within numerical precision under all tested bipartitions.

IV. DISCUSSION

A. Hubs are essential for both classical and quantum connectivity

The monotonic decay of the classical minimal cut confirms the intuitive picture of hubs as the primary bridges holding the causal graph together. The quantum results are far more striking: the complete and robust absence of entanglement under all tested bipartitions demonstrates that hubs are not just one source of quantum correlations, but **the primary source**. Their removal does not merely reduce entanglement; it eliminates it entirely, leaving behind a boundary state with no internal quantum structure. (Recall from Paper II [2] that hubs are nodes with degree ≥ 8 , present in the causal past of every boundary interval.)

B. Complementarity with Paper II: Hubs as the architects of entanglement structure

This result perfectly complements the findings of Paper II. In that work, we showed that **capping** hubs from the start – artificially regularizing the graph’s degree distribution – dramatically increases the entanglement entropy for a specific boundary interval that was chosen to align with the network’s structure. Taken together, the two papers paint a complete picture:

- **Paper II (Hubs Present, Targeted Split):** Hubs generate a rich, localized entanglement structure that can be captured by an appropriately chosen boundary partition.
- **This Paper (Hubs Removed, All Tested Splits):** Without hubs, this structure vanishes, leaving no entanglement for any partition to detect.

Hubs are therefore **both necessary and sufficient** for creating the kind of structured entanglement expected in holographic dualities.

C. Understanding the initial state (Step0)

The inability to contract the full tensor network at step0 (due to nodes with degree > 64) prevents direct measurement of the initial entanglement. However, Paper II [2] provides crucial context: uncapped graphs of similar structure ($N = 2000$) have a hub in the causal past of every boundary interval and exhibit very low entanglement ($R \approx 0.05$ at $\chi = 4$). The classical min-cut at step0 in the present study is 19.3 ± 0.9 , indicating substantial capacity for entanglement even if the actual quantum entropy is modest. Thus the zero entanglement after hub removal represents a genuine loss of correlations that were at least potentially present, not merely a confirmation of a trivial initial state. Moreover, the monotonic decay of the classical cut confirms that the graph’s connectivity—and therefore its capacity to support entanglement—is progressively destroyed.

D. Limitations and future work

- Our study used a modest ensemble of 20 graph realizations. Extending the analysis to a larger ensemble would further strengthen the statistical generality of our conclusions.
- The tensor network contractions, while robust, were limited to bond dimension $\chi = 4$. Higher bond dimensions might reveal residual entanglement not captured here.
- A direct comparison with random node removal was not feasible due to computational constraints (nodes with degree > 64 prevented reliable contractions). Developing more efficient contraction methods or using degree-capped graphs for the comparison could settle whether the effect is uniquely due to hubs.
- A natural next step is to introduce an external radiation system to simulate a full Page curve and further explore the connection to black hole evaporation.

V. CONCLUSION

Together with Papers I and II, this work completes a trilogy establishing the central role of graph regularity in holographic entanglement. Paper I demonstrated that the classical geometry of stochastic causal graphs exhibits the linear scaling characteristic of de Sitter space. Paper II showed that hubs act as entanglement sinks and that

static hub removal (capping) dramatically boosts boundary entanglement. The present paper demonstrates that dynamical hub removal eliminates entanglement entirely, confirming the causal necessity of hubs for quantum correlations. Across the three studies, a consistent picture emerges: graph regularity is the dominant factor enabling holographic entanglement; perfect tensors play only a secondary role. This refines our understanding of what makes constructions like the HaPPY code work—not just perfect tensors, but their placement on regular graphs—and opens the door to dynamical studies of information processing in discrete spacetimes. A natural next step is to introduce an external reservoir to

simulate the Page curve, as suggested in Section 4.4.

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- [1] H. Anderson, Ensemble statistics of linear entanglement scaling in a causal graph model for de sitter holography, submitted to Physical Review D (2026), paper I.
 - [2] H. Anderson, From hubs to holography: How graph regularity unlocks entanglement, submitted to Physical Review

- D (2026), paper II.
- [3] Step0 is omitted because tensor network contraction was not feasible for all graphs due to memory limits with many high-degree nodes; the core result concerns the state after hub removal.